

Sustained excess nutrient intake can result in resistance of target cells to insulin. Compared to normal patterns after a meal, which of the following best describes an effect of insulin resistance?

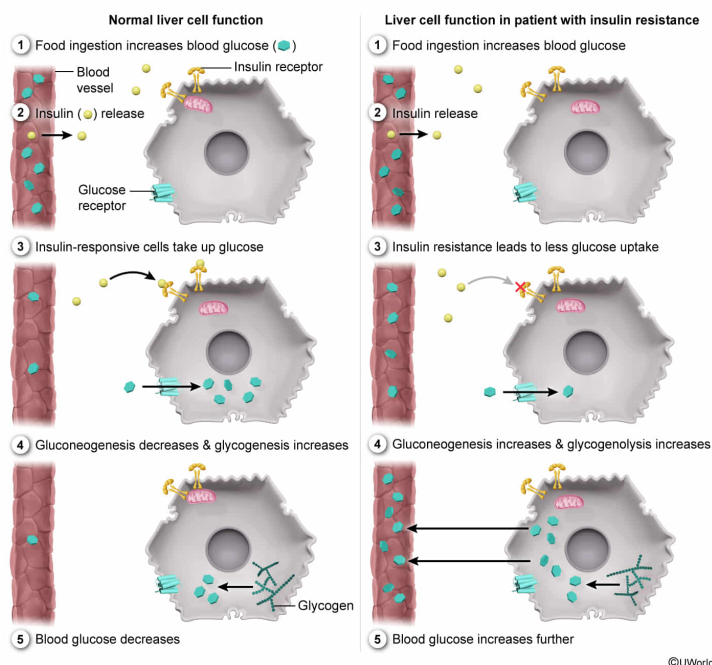
- ☐ A. Liver cells form glycogen at a higher rate [17%]
- ☐ B. Liver cells secrete glucagon at a lower rate [28%]
- ✓ ☒ C. Liver cells produce glucose at a higher rate [47%]
- ☐ D. Kidney cells secrete glucose at a lower rate [6%]

Correct

48% answered correctly

Explanation:

Liver cells increase glucose production in patients with insulin resistance



The liver is responsible for **several functions**, including the regulation of blood glucose, molecule storage (eg, iron, glycogen), synthesis of macromolecules (eg, plasma proteins), **bile production**, and breakdown/detoxification of drugs and waste products (eg, bilirubin).

The control of blood glucose is important in the maintenance of **homeostasis**. **Food ingestion** leads to an increased glucose level and stimulates specialized endocrine cells of the pancreas, called **beta cells**, to release **insulin**. Insulin is a **glucose metabolism hormone** that

decreases the concentration of glucose in the blood. Insulin accomplishes this by *promoting* glucose uptake by tissues (eg, adipose, muscle) and *decreasing* liver glucose production (ie, via [gluconeogenesis](#), [glycogenolysis](#)), while at the same time *promoting* liver glycogen formation (ie, [glycogenesis](#)).

The question states that sustained excess nutrient intake can result in [resistance](#) of target cells to insulin. Liver cells are targeted by insulin to *decrease* glucose production and *increase* glycogen formation after a meal. However, if these target cells are resistant to insulin, glucose production will likely occur at a higher rate than normal and glycogen formation will likely occur at a lower rate than normal. Therefore, compared to normal patterns after a meal, insulin resistance is likely to result in **higher rates of glucose production by liver cells**.

(Choice A) In response to insulin *resistance* (ie, *decreased* glucose uptake by cells), the liver will likely decrease (*not* increase) the rate of glycogen formation.

(Choice B) [Glucagon](#) is a hormone released by the pancreas (*not* liver cells) in response to low blood glucose levels that [opposes](#) the effects of insulin. Release of glucagon results in increased gluconeogenesis and glycogenolysis and in decreased glycogenesis. It is true that the rate of glucagon secretion by the pancreas will likely decrease if a patient is experiencing insulin resistance.

(Choice D) Insulin resistance will result in *high* blood glucose levels. This will likely increase (*not* decrease) the rate of glucose [secretion](#) (ie, into urine) by kidney cells to rid the body of excess glucose.

Educational objective:

In patients with insulin resistance (ie, decreased or absent response of target cells to insulin), the liver will increase production of glucose and decrease production of glycogen after a meal. The liver normally decreases production of glucose and increases production of glycogen after a meal.